Contaminant tracking with Ecotracer
Basic theory and model building

William Walters
Shawn Booth
Villy Christensen

Ecopath 35 Years Conference, St. Petersburg, FL
Dec. 9th 2019

PennState
College of Engineering

Ken and Mary Alice Lindquist Department of Nuclear Engineering

Approximate Schedule

- 1:30-2:15 Introduction and theory
- 2:15-3:00 Hands on example 1
- 3:00-3:30 Break
- 3:30-5:00 Hands on example 2
- 5:00-5:30 Discussion

There is lots of potentially nasty stuff floating around out there

- Polychlorinated biphenyls (PCBs)
- Radionuclides (e.g., cesium, iodine, strontium)
  - Nuclear weapons testing,
- Mercury and heavy metals
- Polycyclic Aromatic Hydrocarbons (oil spills)
- Pharmaceuticals and personal care products
- Microplastics

Concentration ratio method

- A simple and standard method to estimate the concentration of contaminant in organisms

$$CR = \frac{Contaminant\ amount\ in\ organism\ \left(\frac{Bq}{kg}\right)}{Contaminant\ amount\ in\ water\ \left(\frac{Bq}{L}\right)}$$

- Increased amount in the water -> increased amount in organism
- These values are tabulated for many radionuclides and animal species (IAEA)
- Also called bioaccumulation factor for chemical (non-radioisotope) sources
- Assumes an equilibrium situation
  - Not applicable to a nuclear accident or other rapid releases
  - Can not account for changes in consumption or mortality due to other causes

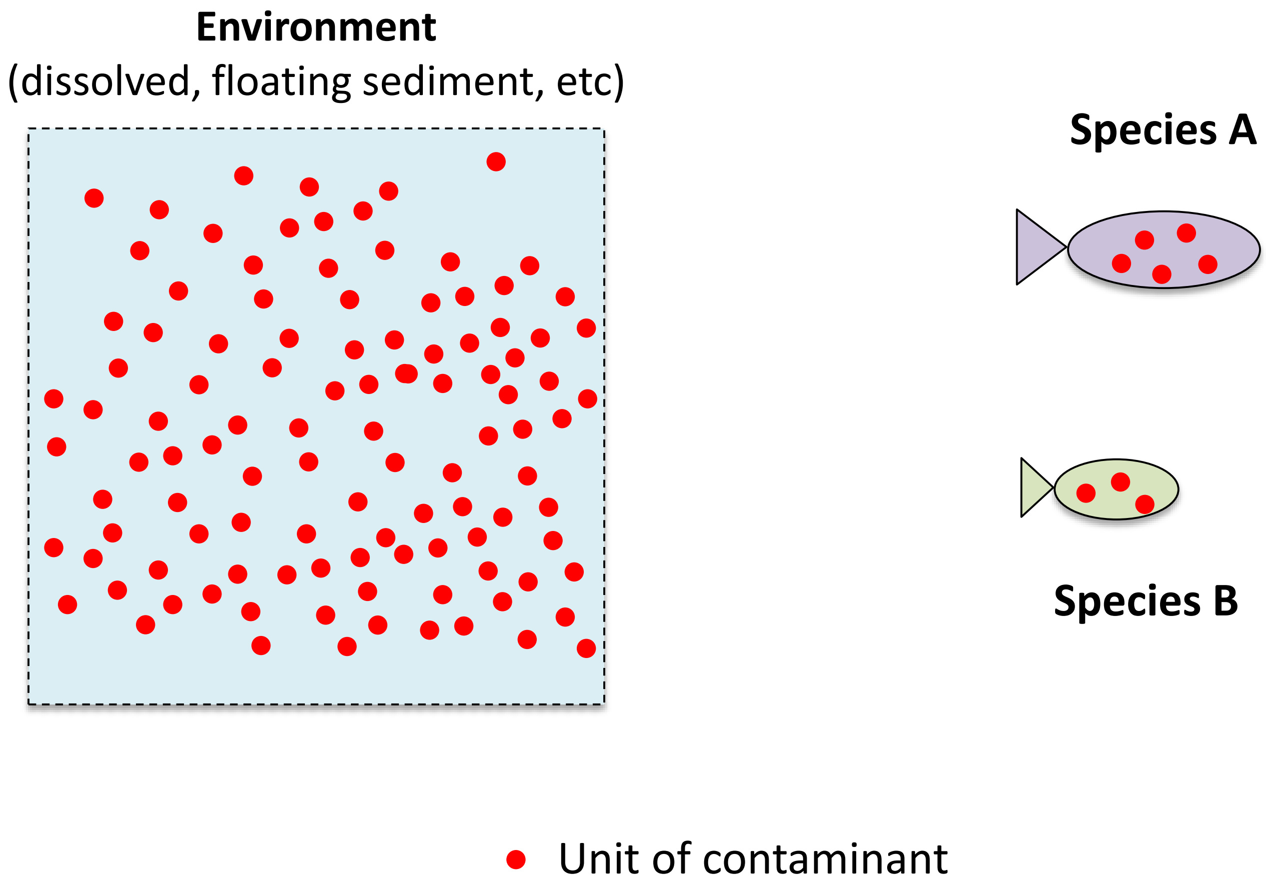
Dynamic methods

- Instead of assuming equilibrium, construct a differential equation

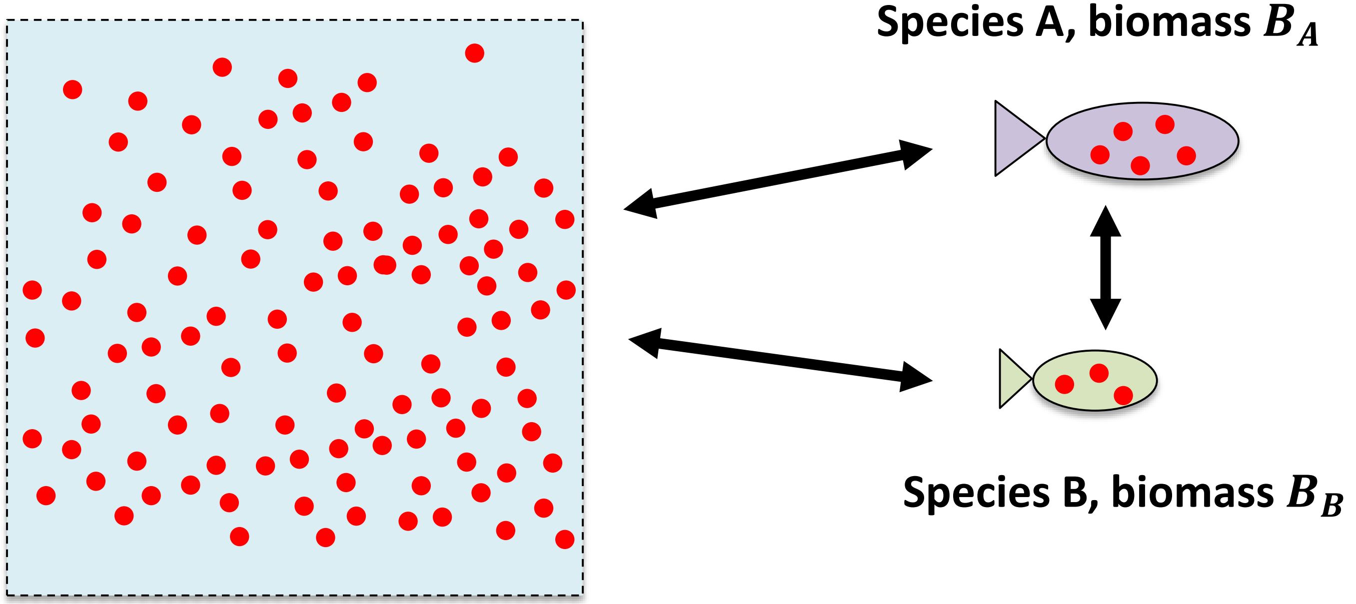
$$\frac{dC_i}{dt} = \alpha - \beta C_i$$

- Parameters can be chosen such that the equilibrium concentrations are the same as with the CR approach
- Models are much more difficult to put together than CR
- Others have used this approach
- Ecotracer has the advantage of integration with Ecopath, Ecosim, Ecospace, and the entire community

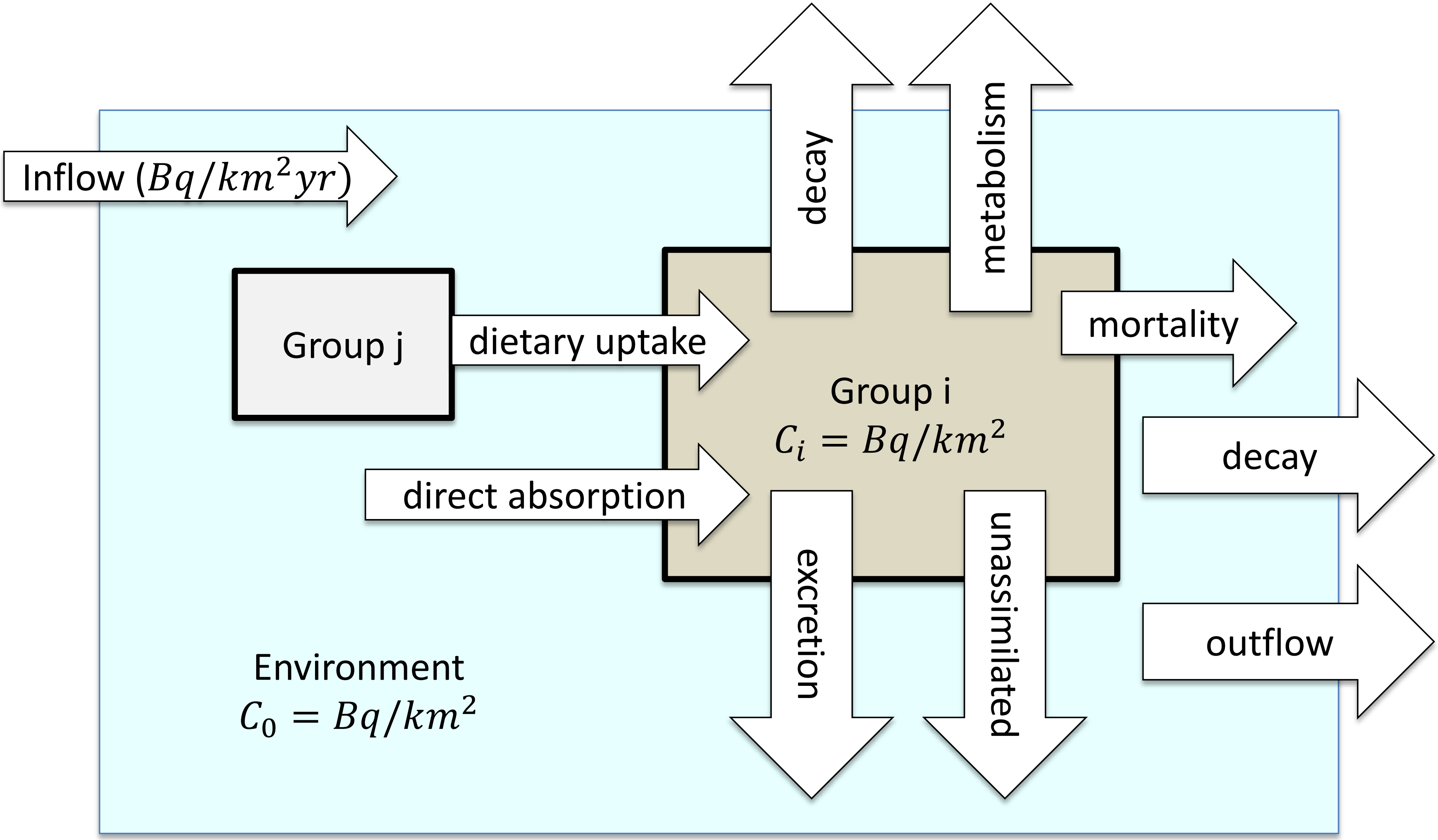
Contaminants exist in “compartments”



Contaminants can be transferred between different compartments



Although contaminants may be very different, there are a few universal mechanisms for these transfers


Inflow (Bq/km^2yr)
decay
metabolism
Group j
dietary uptake
mortality
Group i
$C_i = Bq/km^2$
decay
direct absorption
excretion
unassimilated
outflow
Environment
$C_0 = Bq/km^2$

Units in Ecotracer

- Units can in theory be anything you want, as long as you are consistent. However, you probably should use contaminant values per unit area, so that's it's consistent with the Biomass

$$B_i = t/km^2$$

$$C_i = Bq/km^2 \qquad \text{or} \qquad C_i = g/km^2$$

- Note this is different from standard per volume measurements (e.g., Bq/L)
- Instead of Bq, could use anything else (g, mg, etc)
- This means that the concentration per biomass is:

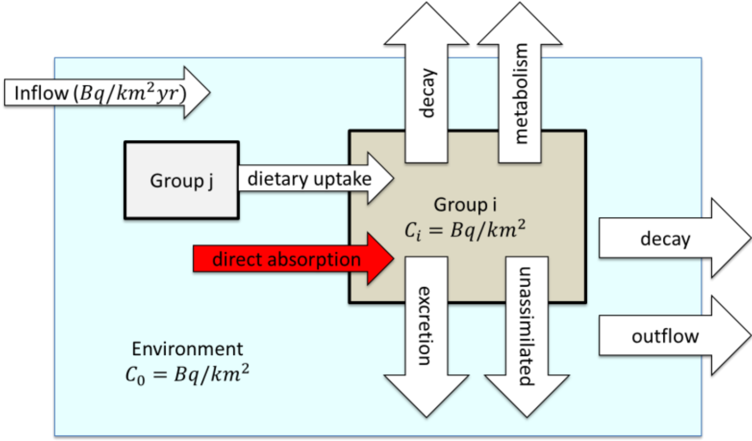
$$\frac{C_i}{B_i} = \left[\frac{Bq}{t}\right] \qquad \text{or} \qquad \frac{C_i}{B_i} = \left[\frac{g}{t}\right] = ppm$$

- The differential equation has

$$\frac{dC_i}{dt} = \alpha - \beta C_i \qquad \text{or,} \qquad \frac{dC_i}{dt} = gains - losses$$

- The terms in the equation should then have units $\left[\frac{Bq}{km^2 \cdot yr}\right]$ or $\left[\frac{g}{km^2 yr}\right]$
- Ecotracer internally has no units – it just multiplies and adds the terms!

Direct uptake



- Direct uptake from the environmental pool C_0

$$u_i B_i C_0$$

  - B_i is biomass of organism i $\left[\frac{t}{km^2}\right]$
  - C_0 is the conc. In the environment $\left[\frac{Bq}{km^2}\right]$
  - u_i is the direct uptake factor $\left[\frac{km^2}{t \cdot yr}\right]$

- This represents a loss from the environment, and a gain to compartment i

Units for environmental uptake

- Lets say we choose:

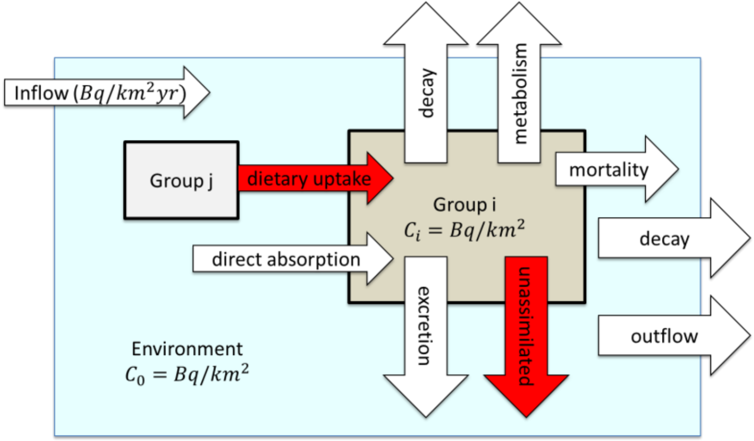
$$B_i = t/km^2$$
$$C_i = Bq/km^2$$

- Then to find the units for u_i, look to the units for the equation

$$\frac{dC_i}{dt}\left[\frac{Bq}{km^2yr}\right] = u_iB_iC_0$$

$$u_i = \frac{\left(\frac{dC_i}{dt}\right)}{B_iC_0} = \frac{\left[\frac{Bq}{km^2yr}\right]}{\left[\frac{t}{km^2}\right]\left[\frac{Bq}{km^2}\right]} = \frac{km^2}{t \cdot yr}$$

Predation uptake term



- Predation is the other gain term for organism i

$$f_i \sum_{j=prey} Q_{ji} \left(\frac{C_j}{B_j} \right)$$

  - Q_{ji} is the consumption rate of prey j by i $\left[\frac{t}{km^2yr}\right]$
  - $\frac{C_j}{B_j}$ is the conc. per biomass of the prey organism j $\left[\frac{Bq}{ton}\right]$
  - f_i is the fraction of contaminant consumed that is retained by organism i
- This predation is a loss from prey organisms j
- The fraction <u>not</u> retained $(1 - f_i)$ is sent to detritus
- Unit check $\left[\frac{t}{km^2yr}\right]\left[\frac{Bq}{t}\right] = \left[\frac{Bq}{km^2yr}\right]$

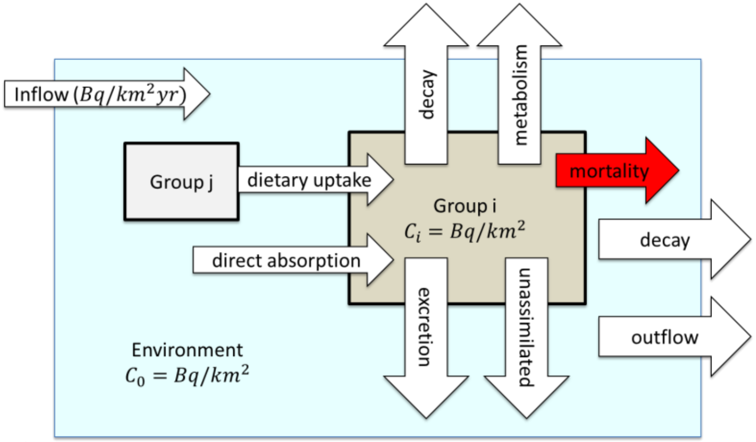
Immigration uptake/Emigration loss

- Contaminant can also be increased by immigration of contaminated species

$$I_i \left(\frac{C_i}{B_i}\right)_{immig}$$

  - I_i is amount of immigration of i $\left[\frac{t}{km^2 yr}\right]$
  - $\left(\frac{C_i}{B_i}\right)_{immig}$ is conc. per biomass of incoming species $\left[\frac{Bq}{ton}\right]$
- Unit check $\left[\frac{t}{km^2 yr}\right]\left[\frac{Bq}{ton}\right] = \left[\frac{Bq}{km^2 yr}\right]$

Mortality loss terms



- All mortality is a loss from the compartment

$$\left(\sum_{j=pred} \frac{Q_{ij}}{B_i} + F_i + MO_i \right) C_i$$

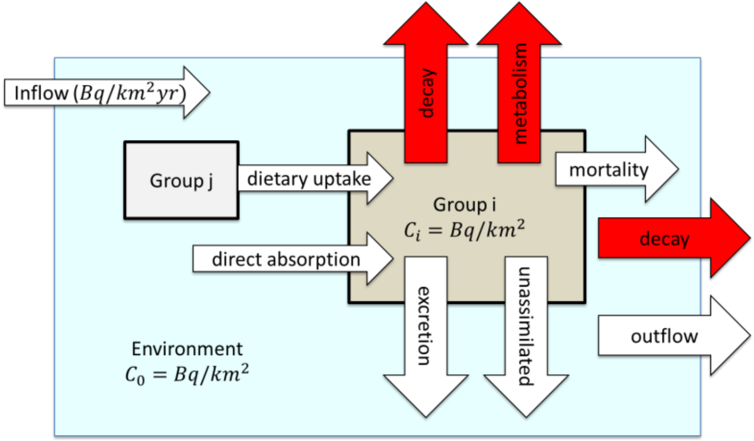
$\frac{Q_{ij}}{B_i}$ predation rate of organism i by j $\left[\frac{1}{yr}\right]$

F_i fishing mortality rate $\left[\frac{1}{yr}\right]$

MO_i other mortality $\left[\frac{1}{yr}\right]$

- Units $\left[\frac{1}{yr}\right]\left[\frac{Bq}{km^2}\right] = \left[\frac{Bq}{km^2yr}\right]$
- Predation loss goes to predator + detritus
- Fishing loss goes out of the system
- Other mortality goes to detritus

Excretion and physical decay



- Biological processes can either
  - excrete the contaminant back to the environment
  - convert it to some inert form (e.g., demethylization), which removes it from the system
- Radioactive or physical decay will also remove the item from the system

$$loss = (m_i + d_i)C_i$$

m_i all loss that go back to the environment (excretion/metabolism) $\left[\frac{1}{yr}\right]$

d_i losses out of system (physical or radioactive) $\left[\frac{1}{yr}\right]$

- Radioactive decay with half-life $t_{1/2}$

$$d_i = \frac{\ln 2}{t_{1/2}}$$

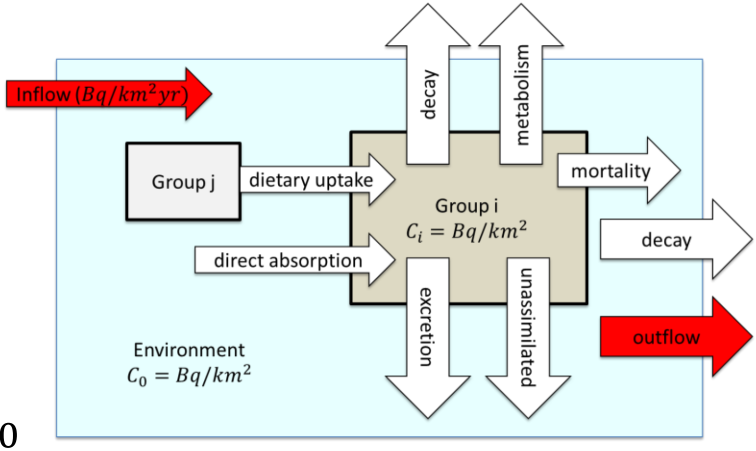
Emigration loss

- Emigration $E_i = \left[\frac{1}{yr}\right]$ is always a loss:

$$E_i C_i$$

- Units $\left[\frac{1}{yr}\right]\left[\frac{Bq}{km^2}\right] = \left[\frac{Bq}{km^2 yr}\right]$

Other environmental terms



- Environmental inflow is a direct addition to C_0
- Outflow depends on the depends on the concentration

$$EI + EO \cdot C_0$$

EI =environmental inflow rate $\left[\frac{Bq}{km^2yr}\right]$

EO = environmental outflow fractional rate $\left[\frac{1}{yr}\right]$

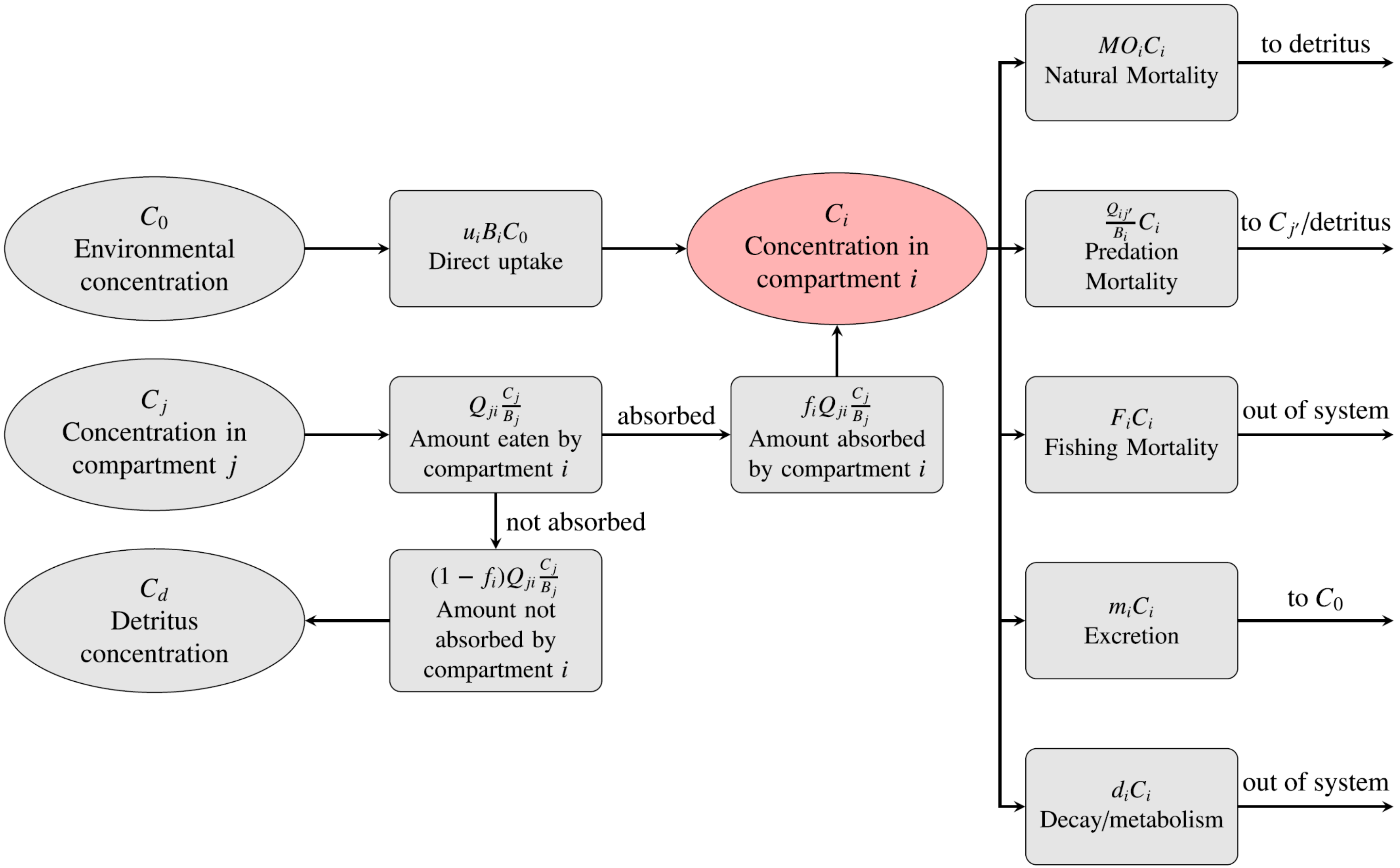
- There is an additional term from the loss of detritus (which is done to conserve overall detritus biomass)

$$D_d C_d$$

D_d = decay/breakdown rate of detritus (calculated internally)

Another remark about units

- Ecotracer has no units or unit conversions in the code
- All terms are calculated exactly as given above



C_0
Environmental concentration
$u_i B_i C_0$
Direct uptake
C_i
Concentration in compartment i
C_j
Concentration in compartment j
$Q_{ji}\frac{C_j}{B_j}$
Amount eaten by compartment i
absorbed
$f_i Q_{ji}\frac{C_j}{B_j}$
Amount absorbed by compartment i
not absorbed
C_d
Detritus concentration
$(1-f_i)Q_{ji}\frac{C_j}{B_j}$
Amount not absorbed by compartment i
$MO_i C_i$
Natural Mortality
to detritus
$\frac{Q_{ij'}}{B_i}C_i$
Predation Mortality
to $C_{j'}$/detritus
$F_i C_i$
Fishing Mortality
out of system
$m_i C_i$
Excretion
to C_0
$d_i C_i$
Decay/metabolism
out of system

Governing equations for Ecotracer

- Put all of these together to create a differential equation

$$\frac{dC_i}{dt} = gains - losses$$

- Ignoring a few terms for brevity:

$$\frac{dC_i}{dt} = u_i B_i C_0 + f_i \sum_{j=prey} Q_{ji} \left(\frac{C_j}{B_j}\right) + - \left(\sum_j Q_{ij} + F_i + MO_i + m_i + d_i\right) C_i$$

- Many of these values are required input for or estimated by Ecopath/Ecosim/Ecospace
  – Biomasses, feeding rates, inflow/outflow (Ecospace)

Parameters for Ecotracer

$$\frac{dC_i}{dt} = \boldsymbol{u_i} B_i \boldsymbol{C_0} + \boldsymbol{f_i} \sum_{j=prey} Q_{ji} \left(\frac{C_j}{B_j}\right) + -\left(\sum_j Q_{ij} + F_i + MO_i + \boldsymbol{m_i} + \boldsymbol{d_i}\right) C_i$$

- Required parameters:
  - u_i direct uptake rates
  - $C_i(0)$ initial concentrations
  - environmental inflow for C_0
  - $(C_i/B_i)_{immig}$ concentration in incoming biomass
  - f_i fraction absorbed
  - m_i excretion rate
  - d_i physical decay rate
- Some of these can be estimated based on laboratory experiments
- Can be chosen such that at equilibrium, the CR values are preserved

All terms together

Biomass pools:

$$\frac{dC_i}{dt} = \boldsymbol{u_i} B_i C_0 + \boldsymbol{f_i} \sum_{j=prey} Q_{ji} \left(\frac{C_j}{B_j}\right) + \boldsymbol{c_i} I_i - \left(\sum_{j=pred} \frac{Q_{ij}}{B_i} + F_i + MO_i + E_i + \boldsymbol{m_i} + \boldsymbol{d_i} \right) C_i$$

direct uptake predation immig. mortality, fishing, emig., excretion, decay

Environment pool

$$\frac{dC_0}{dt} = \boldsymbol{EI} + E_d C_d - \boldsymbol{EO} \cdot C_0 - \sum_i \boldsymbol{u_i} B_i C_0$$

inflow, detr. export, outflow, removal to biota

Detritus pool:

$$\frac{dC_d}{dt} = \sum_{i=pred} (1 - \boldsymbol{f_i}) \sum_{j=prey} Q_{ji} \left(\frac{C_j}{B_j}\right) + \boldsymbol{u_d} B_d C_0 - \sum_{j=pred} \frac{Q_{dj}}{B_d} C_d - E_d C_d$$

non-absorbed predation direct uptake consumed export

Calculating equilibrium values

$$\frac{dC_i}{dt} = gains - losses$$
$$\frac{dC_i}{dt} = \alpha - \beta C_i$$

$$\frac{dC_i}{dt} = u_i B_i C_0 + f_i \sum_{j=prey} Q_{ji}\left(\frac{C_j}{B_j}\right) - \left(\sum_j Q_{ij} + MO_i + m_i + d_i\right) C_i$$

To determine equilibrium values, set $\frac{dC_i}{dt} = 0$ (i.e., not changing)

$$C_{i,eq} = \alpha/\beta$$

Note: C_i is a population-wide value, and each individual fish will likely not ever be at equilibrium

Detritus equilibrium

$$0 = \sum_{i=pred} (1 - \boldsymbol{f_i}) \sum_{j=prey} Q_{ji} \left(\frac{C_j}{B_j}\right) + \boldsymbol{u_d} B_d C_0 - \sum_{j=pred} \frac{Q_{dj}}{B_d} C_d - \frac{E_d C_d}{B_d}$$

$$\frac{C_d}{B_d} = \frac{\left(u_i C_0 + \sum_{i=pred}(1 - \boldsymbol{f_i}) \sum_{j=prey} Q_{ji} \left(\frac{C_j}{B_j}\right)\right)}{\sum_{j=pred} Q_d + E_d}$$

This amount on the bottom is the total biomass removal rate of detritus. Typically, the extra detritus is ‘exported’, in order to keep this biomass constant. Equilibrium, the denominator should just be the total detritus production rate.

Simple example break

- Verify some of the mathematics
- Take a look at outputs, compare to analytical results
  - Steady state
  - Time-dependent

Ken and Mary Alice Lindquist Department of Nuclear Engineering
PennState College of Engineering

Estimating environmental uptake from CR

- Envir. uptake can be estimated based on the CR factor (equilibrium)
- Does not need to be at complete equilibrium but $\frac{dC_i}{dt}$ should be small relative to other terms

$$\frac{dC_i}{dt} = 0 = u_i B_i C_0 + f_i \sum_{j=prey} Q_{ji}\left(\frac{C_j}{B_j}\right) - \left(\sum_{j=pred} \frac{Q_{ij}}{B_i} + MO_i + m_i + d_i\right) C_{i,eq}$$

- At equilibrium, total mortality rate is $Z = P/B$

$$u_i = (Z_i + m_i + d_i)\left(\frac{C_{i,eq}/B_i}{C_0}\right) - \frac{f_i}{B_i} \sum_{j=prey} Q_{ji}\left(\frac{C_j}{B_j C_0}\right)$$

- But we know that the CR value is: $CR_i = \frac{C_{i,eq}/B_i}{C_0}$

$$u_i = (Z_i + m_i + d_i) CR_i - f_i \sum_{j=prey} \frac{Q_{ji}}{B_i} CR_j$$

- Note: this can yield negative u_i if your estimates are not great!
- Units for this CR <u>may</u> be different than standard values!

Estimating environmental uptake

$$0 = u_i B_i C_0 + f_i \sum_{j=prey} Q_{ji} \left(\frac{C_j}{B_j}\right) - (Z_i + m_i + d_i) C_{i,eq}$$

- At equilibrium, total mortality rate is $\mathrm{Z} = P/B$

$$u_i = \frac{losses - cons.\,gains}{B_i C_0}$$

- If you do not know m_i, it's not a big deal
  - No matter your estimate of m_i, you will calculate u_i such that you get the correct equilibrium $C_{i,eq}$

Concentration ratio units

- Always be sure to check the units

$$CR_i = \frac{C_i/B_i}{C_0}$$

- Traditionally this has units $\left[\frac{Bq}{kg}\right] / \left[\frac{Bq}{L}\right]$
- You could define your Ecotracer units so that the CR units work out, or you can change the CR units so that they match Ecotracer

$$CR_i = \frac{\left[\frac{Bq}{km^2}\right] / \left[\frac{t}{km^2}\right]}{\left[\frac{Bq}{km^2}\right]} = \frac{\left[\frac{Bq}{t}\right]}{\left[\frac{Bq}{km^2}\right]} \neq \frac{\left[\frac{Bq}{kg}\right]}{\left[\frac{Bq}{L}\right]}$$

- Not that the Ecotracer units are *per unit area*, not per unit volume. So the depth needs to be taken into account

Concentration ratio and depth

$$CR_i = \frac{\left[\frac{Bq}{km^2}\right] / \left[\frac{t}{km^2}\right]}{\left[\frac{Bq}{km^2}\right]} = \frac{\left[\frac{Bq}{t}\right]}{\left[\frac{Bq}{km^2}\right]} \neq \frac{\left[\frac{Bq}{kg}\right]}{\left[\frac{Bq}{L}\right]}$$

- Not that the Ecotracer units are *per unit area*, not per unit volume. So the depth needs to be taken into account
- Conc per area = Conc per volume * depth

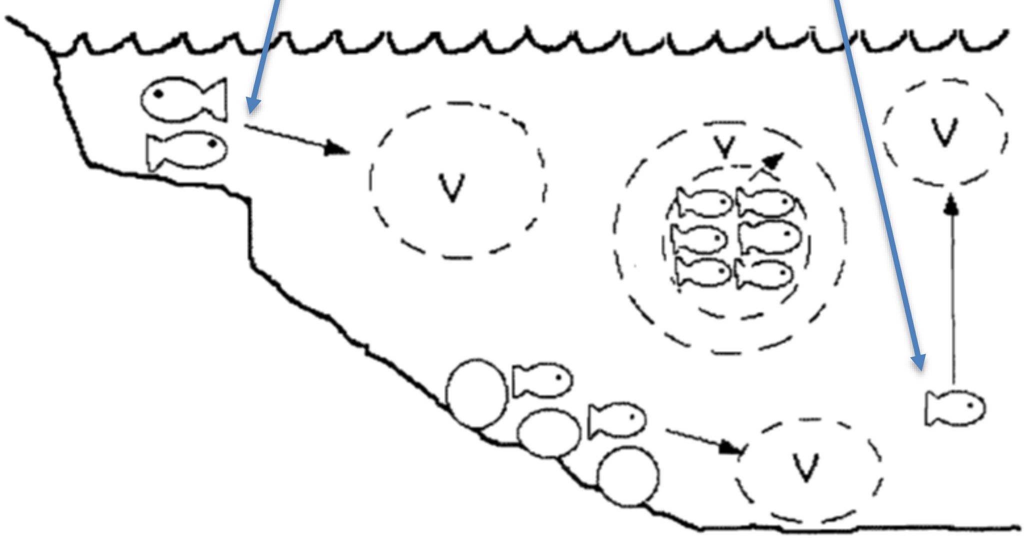
$$\left[\frac{1\ Bq}{L}\right]\left[\frac{1000\ L}{1\ m^3}\right]\left[\frac{1000\ m}{km}\right]^2 \times Depth[m] = \left[\frac{Bq}{km^2}\right]$$

- This assumes that the contaminant is uniformly mixed -> not always a great assumption
  - You can always adjust u_i to account for the higher/lower concentration at the depth the organism resides relative to the average over the entire depth

Water depth and uptake rate

- Ecotracer uses contaminant per unit area $\left[\frac{Bq}{km^2}\right]$ for uptake $uptake = u_i B_i C_0$
- Biological effects based on per unit volume $\left[\frac{Bq}{L}\right]$
- But This can be adjusted by changing the u_i based on depth

Equal per volume, not equal per area!



Variable concentration by depth

- Example:
  - Top layer: 0-10 m, 5.0 $\frac{Bq}{m^3}$
  - Middle later 10-100m, 1.0 $\frac{Bq}{m^3}$
  - Bottom layer 100-120m, 10 $\frac{Bq}{m^3}$

- How much is there in the entire km^2?
  - Top layer: 5.0 $\frac{Bq}{m^3}\left[\frac{1000m}{km}\right]^2[10\ m] = 5 \times 10^7\,\frac{Bq}{km^2}$
  - Middle layer 1.0 $\frac{Bq}{m^3}\left[\frac{1000m}{km}\right]^2[90\ m] = 9 \times 10^7\,\frac{Bq}{km^2}$
  - Bottom Layer 10.0 $\frac{Bq}{m^3}\left[\frac{1000m}{km}\right]^2[20\ m] = 2 \times 10^8\ Bq/km^2$
  - Total $3.4 \times 10^8\ Bq/km^2$

Average per-volume concentration

- Top layer: 0-10 m, 5.0 $\frac{Bq}{m^3}$
- Middle later 10-100m, 1.0 $\frac{Bq}{m^3}$
- Bottom layer 100-120m, 10 $\frac{Bq}{m^3}$

- Total $3.4 \times 10^8\ Bq/km^2$
- Average per volume : $\frac{\left(3.4\times10^8 \frac{Bq}{km^2}\right)}{120m}\left[\frac{1\ km}{1000\ m}\right]^2 = 2.83\,\frac{Bq}{m^3}$

- In Ecotracer, the depth is all averaged out
  - Species in higher conc. depths need to have their u_i increased to compensate
  - Opposite effect in lower conc. regions

Example for depth calculating u_i

- Lets say an organism only has environmental uptake, and has

$$CR_i = 50\frac{\left[\frac{Bq}{kg}\right]}{\left[\frac{Bq}{m^3}\right]} = 50\frac{m^3}{kg}$$

Converted to Ecotracer units, this is

$$CR_i^{(EwE)} = 50\frac{m^3}{kg}\left[\frac{1000\text{ kg}}{\text{t}}\right]\left[\frac{km}{1000\,m}\right]^2\left[\frac{1}{120\,m}\right] = 4.16\times10^{-4}\frac{km^2}{t}$$

In the bottom layer, we would expect to have:

$$\frac{C_i}{B_i} = CR_i\,C_0^{(actual)} = 50\frac{m^3}{kg}\cdot 10\frac{Bq}{m^3} = 500\frac{Bq}{kg} = 500{,}000\frac{Bq}{t}$$

Using EwE units we have instead:

$$\frac{C_i}{B_i} = CR^{(EwE)}C_0 = 4.16\times10^{-4}\frac{km^2}{t}\cdot 3.4\times10^8\frac{Bq}{km^2} = 142{,}000\frac{Bq}{t}$$

Example for depth calculating u_i

We notice that the factor $\frac{500{,}000\,\frac{Bq}{t}}{142{,}000\,\frac{Bq}{t}} = 3.53$

Is exactly the same as the local-to-average depth ratio:

$$\frac{10\,\frac{Bq}{m^3}}{2.83\,\frac{Bq}{m^3}} = 3.53$$

- Therefore, you can compensate for a non-uniform contaminant depth profile by adjusting the u_i by the local-to-average depth ratio
- The environmental uptake is the only parameter than should be affected by this issue
- Note: this should not affect calculating u_i if you already know the concentration. But if you <u>know</u> u_i for a species, then you can use this to apply to different scenarios

Model building practices

- If your model has known equilibrium concentrations, you should be able to reproduce the equilibrium in Ecotracer
  - It does take a bit of work
  - Maybe an Ecotracer-path type tool would be useful for future model building
- Start simple, verify that units are working out
- Start at lowest trophic levels and verify them first
  - primary producers are very straightforward
  - no diet terms
  - not affected by the concentrations of ‘higher up’ organisms